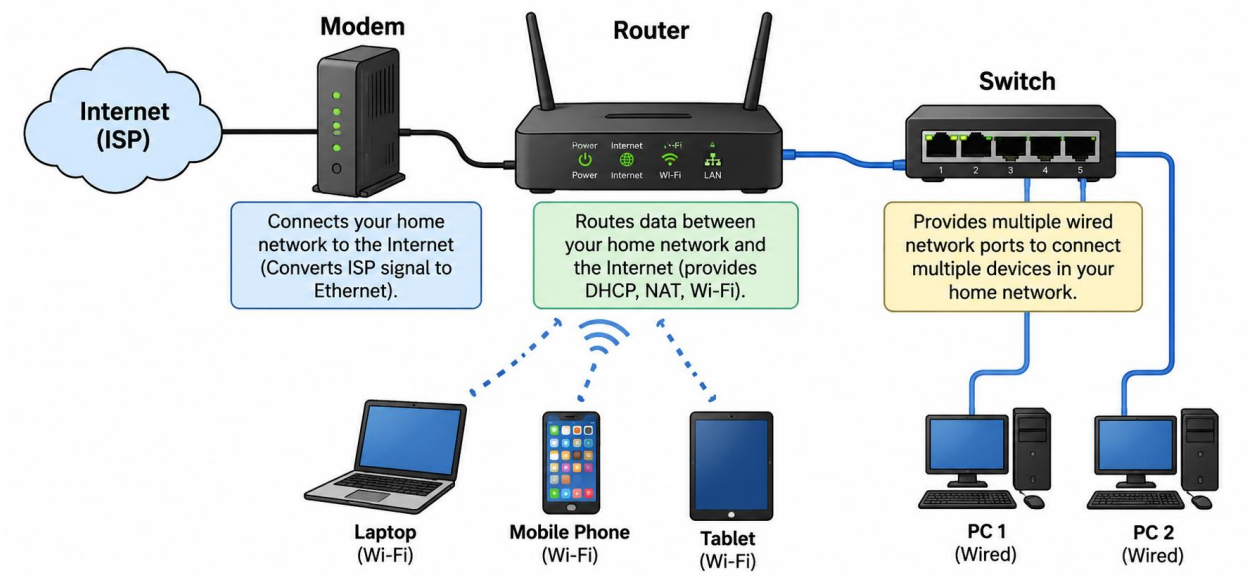
Routers/modems/switches

Tasks :-

1.Draw a simple network diagram showing how a router, modem, and switch connect together in a typical home setup. Label each device and explain the role of each in one line on your diagram.

Ans :-



**2.Log in to your home WiFi router's admin panel (usually at 192.168.0.1 or 192.168.1.1) and find the section that shows connected devices. List the names or MAC addresses of at least 3 devices currently connected to your network.

Hint: If you don't have access to a physical router, search online for a router admin demo page and use that for this task.**

Ans :-

- **Connected Devices Table**

No.	Device Name	IP Address	MAC Address	Connection Type
1	Laptop	192.168.1.101	AA:BB:CC:11:22:33	Wi-Fi
2	Android-Phone	192.168.1.102	AA:BB:CC:44:55:66	Wi-Fi
3	Smart-TV	192.168.1.103	AA:BB:CC:77:88:99	Wi-Fi

- The router's connected-device list helps identify devices using the home network through their device names, IP addresses, and MAC addresses. This information can help monitor network usage and identify unfamiliar devices.

3.Create a table comparing routers, modems, and switches. Include at least 4 points for each: main function, where it is used (home/office), whether it assigns IP addresses, and if it supports WiFi.

Ans :-

Comparison Point	Router	Modem	Switch
Main Function	Connects different networks and routes data packets.	Connects the home or office network to the ISP network.	Connects multiple devices within a LAN.
Where Used	Homes, offices, and organizations.	Homes and offices.	Offices, schools, data centers, and LANs.
Assigns IP Addresses	Yes, typically through DHCP.	Usually no; it may receive a WAN IP from the ISP.	No, a basic Layer 2 switch does not assign IP addresses.
Wi-Fi Support	Yes, if it is a wireless router.	Usually no, unless it is a modem-router combo.	No, a standard switch does not provide Wi-Fi.
OSI Layer	Layer 3 – Network Layer.	Primarily Physical and Data Link functions, depending on technology.	Layer 2 – Data Link Layer.
Example	TP-Link, D-Link, Cisco Router.	DOCSIS Cable Modem, DSL Modem.	Cisco Catalyst, TP-Link Ethernet Switch.

**4.Imagine you are setting up a small gaming café. Choose and specify which devices (router, modem, switch) you would need, how many of each, and briefly explain the reason for each choice.

Hint: Consider the number of computers, internet connection, and need for wired vs wireless access.**

Ans :-

- I am setting up a small gaming café with 10 gaming computers, 1 billing computer, and Wi-Fi access for customers and staff. The network should provide stable internet connectivity, low latency, and reliable data transfer for online gaming.

- **Required Network Devices**

No.	Device	Quantity	Reason for Selection
1	Modem / ISP ONT	1	Connects the gaming café to the Internet Service Provider (ISP).
2	Router	1	Manages network traffic, assigns IP addresses using DHCP, and provides Wi-Fi.
3	Gigabit Ethernet Switch (16-Port)	1	Connects all 10 gaming PCs, billing PC, router, and additional wired devices.
4	Wi-Fi Access Point	1	Provides wireless connectivity for customers and staff, if the router's Wi-Fi coverage is insufficient.

- **Why Wired Connections for Gaming?**

- Ethernet provides stable connectivity for gaming computers.

- Wired connections generally have lower and more consistent latency than Wi-Fi.

- A Gigabit switch supports high-speed communication between connected devices.

- Wi-Fi is suitable for mobile phones and casual browsing.